

Omar Mohamed Ibrahim

AI & Data Scientist



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SUMMARY

Recent Computer Science graduate & AI/Data Scientist specializing in scalable backend architectures and integrating Generative AI (LLMs/VLMs) into real-time applications. Adept at building end-to-end ML pipelines—from edge data ingestion to cloud inference under strict latency constraints. Seeking AI Engineer & Data Scientist roles.

TECHNICAL SKILLS

- **Programming & Databases:** Python, SQL (MySQL, MS SQL Server), R
- **Machine Learning & AI:** Deep Learning, Predictive Modeling, Feature Engineering, Regression, Classification, CNN, RNN, LSTM, Transfer Learning
- **Computer Vision & Generative AI:** Object Detection (YOLO), Image Segmentation, LLM Integration, Prompt Engineering, Sentiment Analysis
- **Libraries & Frameworks:** TensorFlow, scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, BeautifulSoup, Selenium
- **Deployment & Tools:** FastAPI, Flask, Streamlit, Docker, Hugging Face, Git
- **Data Analytics & BI:** Data Wrangling, ETL Pipelines, Power BI, Tableau

EDUCATION

Bachelor's Degree in Computer and Information Technology - Future University in Egypt
September 2022 – July 2026

- **Specialization:** Computer Science | **GPA :** 3.1 (Very Good)
- **Graduation Project: Sentinel AI** – Architected and developed an end-to-end automated traffic accident detection and reporting system using multi-sensor fusion, MobileNet + LSTM, and Gemini Flash.
[\[Live Demo\]](#)

PROJECTS

Brain Tumor Detection System | Deep Learning (CNN) & Computer Vision

- Developed an end-to-end diagnostic web application using TensorFlow (CNN) and Flask to classify MRI scans with 98% accuracy.
- Prioritized medical safety by optimizing for 100% Recall to eliminate false negatives.
- Deployed the model via a user-friendly interface allowing real-time image upload and inference.

Smart Mask Detection & Robotic Arm System | Computer Vision & IoT

- Developed a real-time mask detection pipeline using OpenCV and CNNs, achieving 95%+ accuracy in identifying unmasked individuals.
- Integrated vision model with a robotic arm hardware setup to automatically pick and deliver face masks to non-compliant users.
- Optimized inference performance to process live video feeds with under 50ms latency for seamless hardware interaction.

Global Education & Economy Analysis | EDA & Data Analytics (ETL)

- Built a unified global dataset (10,000+ records) by scraping, cleaning, and merging 5+ disparate sources to perform Exploratory Data Analysis (EDA) via Python (Pandas).
- Derived actionable insights on literacy-to-GDP correlations across 190+ countries, automating data prep to reduce wrangling time by 30%.
- Built interactive dashboards using Streamlit & Plotly to visualize economic trends for multi-country comparisons.

EXPERIENCE

Data Analysis Trainee - AITB (July 2025 -Aug 2025)

- **Exploratory Data Analysis (EDA) & KPI Tracking:** Processed, cleaned, and analyzed 10+ complex datasets (100k+ records) using Python (Pandas, NumPy) to track key KPIs and business metrics.
- **Data Visualization & Dashboarding:** Built 5+ daily-refreshed, automated dashboards using Power BI, Matplotlib, and Seaborn, improving reporting turnaround time for stakeholders by 25%.
- **Workflow Automation:** Developed automated Python scripts for data manipulation and cleaning, reducing manual pipeline processing time by 30%.

CERTIFICATES

- **AMIT** AI & Data Science Diploma
- **Google** Share Data Through the Art of Visualization
- **Google** Data Analysis with R Programming
- **Google** Data Analytics Capstone: Complete a Case Study

LANGUAGES

- **Arabic:** Native.
- **English:** Advanced (C1) – Professional Working Proficiency